From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA INTEGRATED TEST

FIITJEE

JEE(Advanced)-2014

ANSWERS, HINTS & SOLUTIONS

PART TEST – IV

(Paper – 1)

ANSWERS KEY			
	PHYSICS	CHEMISTRY	MATHEMATICS
Q. No.	ANSWER	ANSWER	ANSWER
1.	D	D	C
2.	A	C	C
3.	B	C	B
4.	C	B	D
5.	D	B	B
6.	D	B	B
7.	A	C	A
8.	B	A	B
9.	D	C	A
10.	D	C	C
11.	A, B, C	A, B	A, D
12.	A, B, C	A, B	A, B, C, D
13.	A, B, D	A, C	B, D
14.	A, C, D	A, B	A, C
15.	A, D	A, B	A, B, C, D
1.	4	2	2
2.	2	8	2
3.	3	2	2
4.	2	4	2
5.	5	4	0

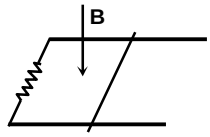
Physics

PART – I

1. $\Delta\phi = \frac{2\pi}{\lambda} d\sin\theta = \frac{10\pi}{3}$

$I = I_0 + I_0 + 2I_0\cos[2\pi + (4\pi/3)] = I_0$

4. $F\cos\theta = \frac{B^2a^2v}{R} + \mu(mg - F\sin\theta)$



$F(\cos\theta + \mu\sin\theta) = \frac{B^2a^2v}{R} + \mu mg$

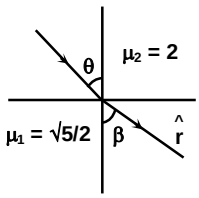
For F to be minimum, $\tan\theta = \mu$

$F_{min} = \frac{\mu mg}{\sqrt{\mu^2+1}} + \frac{B^2a^2}{R}\frac{v}{\sqrt{\mu^2+1}}$

6. $-(\hat{i} - 2\hat{j})\hat{j} = |\hat{i} - 2\hat{j}||\hat{i}|\cos\theta$

$\Rightarrow \sin\theta = 1/\sqrt{5}$

$2 \times \frac{1}{\sqrt{5}} = \frac{\sqrt{5}}{2}\sin\beta \Rightarrow \sin\beta = 4/5$

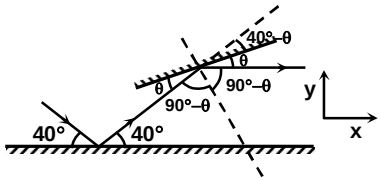


$\Rightarrow \cos\beta = 3/5 \Rightarrow \hat{r} = \frac{4}{5}\hat{i} - \frac{3}{5}\hat{j}$

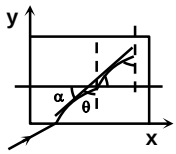
7. $\theta = 40° - \theta \Rightarrow \theta = 20$

$\Rightarrow$ 5° clockwise or 95 clockwise.

In this case the ray will become horizontal and will be reflected along $-\hat{i}$



8. In both C and D μ increases a y is increases. In only B μ decreases as x increased.



9. $f_{lens} = \frac{R}{\mu - 1}$ = 60 cm

10. $\beta = \lambda D/d \Rightarrow \beta \propto \lambda$

The position of the central maxima is independent of wavelength used.

$\gamma_n = \frac{n\lambda D}{d} \quad \therefore \quad \gamma_n \propto \lambda$

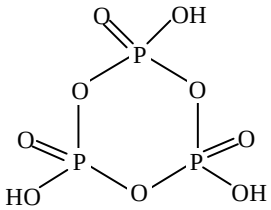
Section –C

1. $\mu = \left(\frac{qBr}{mg}\right)t = 4$

Chemistry

PART – II

SECTION – A

2. 
There are three P–O–P bonds.]

3. Hardy–Schulze rule

4. For $Zn + 2Fe^{+2} \rightarrow Zn^{+2} + 2Fe$ $E^{o}_{cell} = -0.44 + 0.76 = +ve$ and hence, the reaction is spontaneous.

5. Lowering, $\Delta P \propto$ 'a' number of solute particles.
Hence, HA_1 is stronger acid than HA_2 (Given $\Delta P_1 > \Delta P_2$) and $pK_{a_1} < pK_{a_2}$.

6. $TiO_2 \longrightarrow TiCl_4$
$(+4)$ $(+4)$
No change in oxidation number takes place.

7. (X) is KI.
$I^- + Ag^+ \longrightarrow AgI$
(Yellow ppt.)
(insoluble in NH_4OH)
$CuSO_4 + 2KI \longrightarrow CuI_2 + K_2SO_4$
(X)
$2CuI_2 \longrightarrow Cu_2I_2 \downarrow + I_2$
Brown ppt.
$Na_2S_2O_3 + I_2 \longrightarrow Na_2S_4O_6 + 2NaI$
Hence, (C).

8. Milliequivalents of NaOH = 28.5 × 0.125 = 3.5625 milliequivalents
Milliequivalents of NaOH $\equiv$ Milliequivalents of salt
$$3.5625 = \frac{0.319}{[52 + 6 \times 18 + 35.5 \times 3] \times 1000 \times \text{charge}}$$
Charge = +3
So, formula $= [Cr(H_2O)_6]^{3+} + 3Cl^-$, i.e. $[Cr(H_2O)_6]Cl_3$

10. $4Fe^{3+} + K_4[Fe(CN)_6] \longrightarrow Fe_4[Fe(CN)_6]_3$
Prussian blue

11. $P_{total} = P^0_B X_B + P^0_T X_T$

$= 42 \times \frac{1}{3} + 36 \times \frac{2}{3} = 38$ mm.

$P_M Y_B = P_B^0 \times X_B$; Where Y_B is the mole fraction of Benzene is vapour phase.

$38 \times Y_B = 14$

$Y_B = \frac{14}{38} = \frac{7}{19}$.

12. In the given reaction 'X' can be Au and Ag_2S.

$4Au + 8CN^- + 2H_2O + O_2 \longrightarrow 4Au(CN)_2^- + 4OH^-$

$2Ag_2S + 8CN^- + 2H_2O + O_2 \longrightarrow 4Ag(CN)_2^- + 4OH^- + 2S$

13. $Al^{+3} + 3e^- \rightarrow Al$

Number of moles of Al produced = $\frac{4.5}{27} = 0.166$

Number of moles of electrons used = 3×0.166

$H^+ + e^- \rightarrow \frac{1}{2}H_2$

Number of moles of H_2 produced = $\frac{1}{2} \times 3 \times 0.166$

Volume of H_2 produced = $1.5 \times 0.166 \times 22.4$ *l* = 5.57 = 5.6 *l*

15. Rochelle salt (sodium potassium tartrate) and bleaching powder (calcium chloride hypochlorite) are mixed salts. Calcium hypochlorite is a simple salt. Mohr's salt [$FeSO_4.(NH_4)_2SO_4.6H_2O$] is a double salt.

SECTION – C

1. $0.7 = 2 \times K_f \times 0.2 \Rightarrow K_f = \frac{0.7}{0.4} = \frac{7}{4}$

$Hg(CN)_2 + mCN^- \longrightarrow Hg(CN)_{m+2}^{m-}$

0.1	0.2	0
0	(0.2 – 0.1m)	0.1

Now, Molarity of K^+ = 0.2 M

Molarity of CN^- = (0.2 – 0.1m) M

Molarity of complex = 0.1 M

$0.53 = K_f (0.2 + 0.2 – 0.1m + 0.1) = K_f (0.5 – 0.1m)$

$\Rightarrow 0.53 = \frac{7}{4} (0.5 – 0.1m) \Rightarrow 2.1 = 3.5 – 0.7m$

$\Rightarrow 0.7m = 1.4 \Rightarrow m =$ **2**.

2. Number of moles of hypo = $\frac{MV}{1000} = \frac{0.1 \times 40}{1000} = 4 \times 10^{-3}$

$\therefore$ Number of moles of $I_2 = 2 \times 10^{-3}$

Mass of $I_2 = 2 \times 10^{-3} \times 254$ g

$$w = \frac{i \times t \times E}{96500}$$

$$2 \times 10^{-3} \times 254 = \frac{i \times 9650 \times 127}{96500}$$

i = 0.04 ampere

Current efficiency = $\frac{0.04}{0.50} \times 100$ = **8%**.

5. $Fe^{2+} + 2e^- \longrightarrow Fe(s); E^0 = 8V$ ...(1)

$Fe^{3+} + e^- \longrightarrow Fe^{2+}; \quad E^0 = 2V$ ...(2)

Equation (1) + (2):

$Fe^{3+} + 3e^- \longrightarrow Fe(s); E^0 = xV(let)$

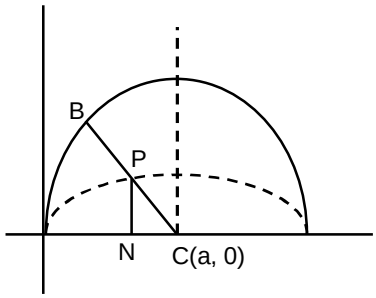
Hence, $3x = 2 \times 5 + 2$

$X = 4$

Mathematics PART – III

1. E: Indian is seated adjacent to his wife
A: American are seated adjacent to their wives
$P\left(\frac{E}{A}\right) = \frac{2}{5}$

2. $y = \sqrt{2ax - x^2} \Rightarrow (x - a)^2 + y^2 = a^2$
Let P(h, k) be a point then BP > PN
For the boundary condition BP = PN = k
Now $AP = a - k = \sqrt{(h-a)^2 + k^2} \Rightarrow k = h - \frac{h^2}{2a}$
$\therefore$ boundary of the region is $y = x - \frac{x^2}{2a}$
required area $= 2\int_0^a \left(x - \frac{x^2}{2a}\right)dx = \frac{2a^2}{3}$.



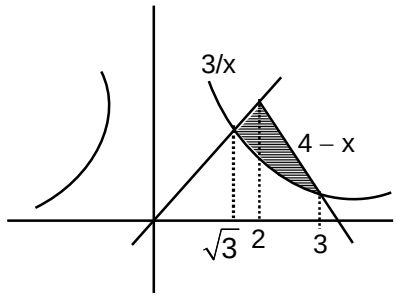
3. $$\frac{\frac{m-1}{m}\times\frac{1}{m}}{\frac{1}{m}\times 1+\frac{m-1}{m}\times\frac{1}{m}} = \frac{m-1}{2m-1}$$

4. $QR = 2\sqrt{2} \Rightarrow OP = 6$.

5. Let $\vec{r} = \alpha\vec{a} + \beta\vec{b} + \lambda(\vec{a}\times\vec{b})$
$\Rightarrow 4\alpha + \beta = 2$
also $\alpha + \beta = 8$
$\therefore \alpha = -2, \beta = 10$
also $\lambda = 2$
now $\vec{r} + 2\vec{a} - 10\vec{b} = 2(\vec{a}\times\vec{b}) = 2$.

6. Applying $R_1 \to R_1 - R_3$ and $2q = p + r$
$$\begin{vmatrix} 2^{n+1} - 2^n + p & 2^{n+2} - 2^{n+1} + q & p + r \\ 2^n + p & 2^{n+1} + q & p + r \\ a^2 + 2^n + p & b^2 + 2^{n+1} + 2q & c^2 - r \end{vmatrix} = \begin{vmatrix} 2^n + p & 2^{n+1} + q & p + r \\ 2^n + p & 2^{n+1} + q & p + r \\ a^2 + 2^n + p & b^2 + 2^{n+1} + 2q & c^2 - r \end{vmatrix} = 0.$$

7. Area $= \int_{\sqrt{3}}^{2} x - \frac{3}{x} + \int_2^3 \left(4 - x - \frac{3}{x}\right)dx = \frac{4 - \ln 27}{2}$.



8. 20 cards are required for 2 spades means in 19 draws there is only one spade and 20th card is a spade.
$\Rightarrow$ The order 18 cards are drawn from non–spades card and 1 card is drawn from 13 spades.
$\therefore$ Required probability $= \frac{^{39}C_{18} \cdot ^{13}C_1}{^{52}C_{19}} \cdot \frac{12}{33}$.

9. Let the equation of plane passing through origin is $lx + my + nz = 0$
where l, m, n are direction cosines, then $l^2 + m^2 + n^2 = 1$ …(i)
The given line is parallel to the plane, then $2l - m - 2n = 0$ …(ii)
The distance between line and plane is 5/3, then $l - 3m - n = 5/3$ …(iii)
From (i), (ii) and (iii), we get equation of plane are
$x - 2y + 2z = 0$ and $2x + 2y - z = 0$.

10. Applying $R_3 \to R_3 - xR_1 - yR_2$

$$\Delta = \begin{vmatrix} p & q & px+qy \\ q & r & qx+ry \\ 0 & 0 & -\left(px^2+2qxy+ry^2\right) \end{vmatrix}$$

$(q^2 - pr)(px^2 + 2qxy + ry^2) < 0$.

11. $2P(x = 5) = P(x = 4) + P(x = 6)$

$2 = \frac{^nC_4}{^nC_5} + \frac{^nC_6}{^nC_5} \Rightarrow n = 7, 14$

12. Since all planes are parallel

$P_1 = \frac{4}{\sqrt{29}}$, $p_2 = \frac{1}{2\sqrt{29}}$, $p_3 = \frac{8}{\sqrt{29}}$

13. Given lines are perpendicular if $2\vec{i} + p\vec{j} + 5\vec{k}$ and $3\vec{i} - p\vec{j} + p\vec{k}$ are perpendicular.
$\Rightarrow 2.3 + p(-p) + 5p = 0 \Rightarrow p = -1, 6$

14. Let $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$. Then $|\vec{c}| = \sqrt{2} = \sqrt{c_1^2 + c_2^2 + c_3^2}$ (1)

And $= \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|} = \frac{1}{2} \Rightarrow \frac{1}{2} = \frac{\vec{a}\cdot\vec{c}}{|\vec{a}||\vec{c}|} = \frac{c_1 + c_2}{\sqrt{2}.\sqrt{2}} \Rightarrow c_1 + c_2 = 1$

and $\frac{1}{2} = \frac{\vec{b}.\vec{c}}{|\vec{b}||\vec{c}|} = \frac{c_2 + c_3}{2} \Rightarrow c_2 + c_3 = 1$

From (1)
$2 = (1 - c_2)^2 + c_2^2 + (1 - c_2)^2$
$\Rightarrow 3c_2^2 - 4c_2 = 0$
$\Rightarrow c_2 = 0$ or $c_2 = \frac{4}{3}$

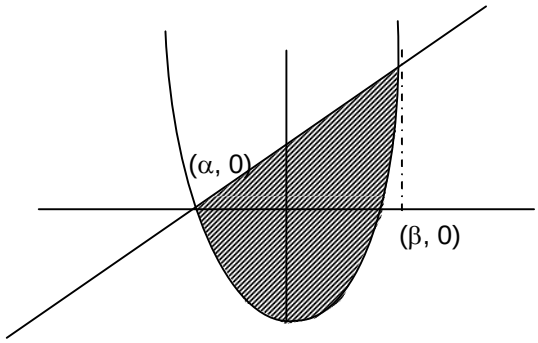
Therefore, the points are (1, 0, 1) and $\left(-\frac{1}{3}, \frac{4}{3}, -\frac{1}{3}\right)$

SECTION – C

1. $A = \int_{\alpha}^{\beta} (y_1 - y_2)\, dx$ where α, β are the roots of $x^2 + 2x - 3 = mx + 1$, on solving we will get

$\frac{1}{6}(m^2 - 4m + 20)^{3/2}$: Hence $\alpha = \frac{32}{3}$

$\Rightarrow \frac{6\alpha}{32} = 2.$



2. E = even of any one cutting a spade in one cut

$n(E) = {}^{13}C_1$

$n(S) = {}^{52}C_1$

$P(E) = 1/4 = p$, $P(\bar{E}) = q$

Probability of a winning = p + qqqp + qqqqqqp+ ……∞

$= \frac{p}{1-q^3} = \frac{64}{175}$

$\lambda = 350 \times \frac{64}{175} = 128$

$\left[\frac{\lambda}{64}\right] = 2$

4. Let a = 1 – x

b = x – y

c = y – z

d = z

then a + b + c + d = 1 and $a^2 + b^2 + c^2 + d^2 = \frac{1}{4}$

$\Rightarrow (a-b)^2 + (a-c)^2 + (a-d)^2 + (b-c)^2 + (b-d)^2 + (c-d)^2 = 0$

$\Rightarrow a = b = c = d$

$\therefore x = \frac{3}{4}, y = \frac{1}{2}, z = \frac{1}{4}$

So the distance from the point $\left(\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right)$

from the plane 4x + 2y + 4z + 7 = 0 is $\frac{3+1+1+7}{6} = 2$

5. $|A - I| = |A - I||A'|$

$|AA' - A|$ $(|AB| = |A|\,|B|)$

$|I - A'| = -|A' - I| = -|A - I|' = -|A - I|$

$\Rightarrow |A - I| = 0$